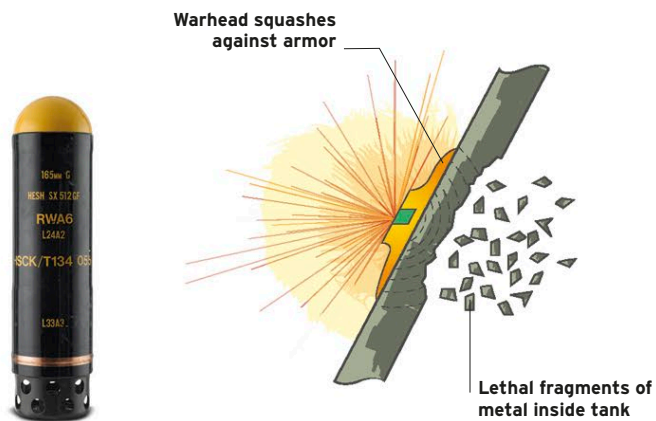


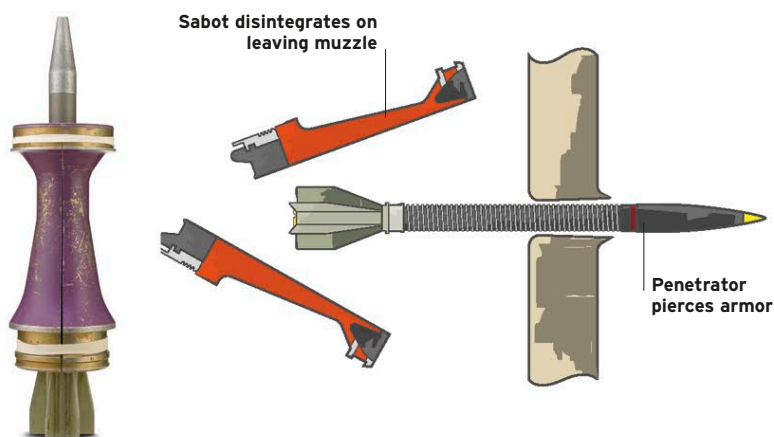
HIGH-EXPLOSIVE SQUASH HEAD

Developed in Britain in the late 1940s, HESH rounds have a very short delay in their fuse. This gives them time to expand across the surface of the armor on impact before detonation. Their explosive force causes partial disintegration of the plate, which drives lethal fragments of metal off the inner surface of the armor, potentially killing crewmen inside the tank.



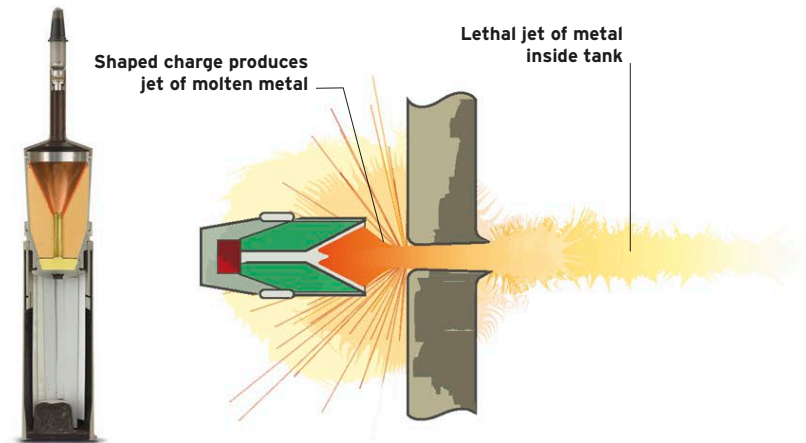
ARMOR-PIERCING FIN-STABILIZED DISCARDING SABOT

APFSDS rounds are the most effective antitank weapons on the modern battlefield. The penetrator dart is made from a highly dense material, often tungsten or depleted uranium, as this maximizes its mass and therefore armor penetration. APFSDS rounds do not spin, since this reduces armor penetration, instead relying on their fins for stability in flight.



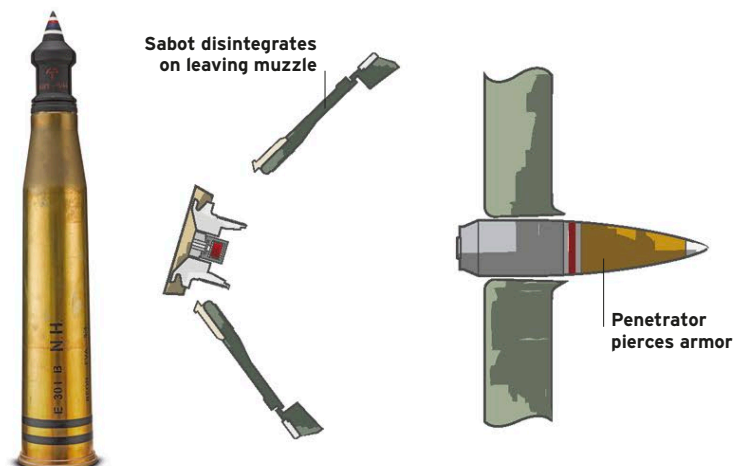
HIGH-EXPLOSIVE ANTITANK

HEAT rounds utilize a shaped charge to produce a "superplastic" jet of molten metal that punches its way through armor plate. It does not burn through: the effect is caused exclusively by kinetic energy. This Munroe effect, as it is also known, is widely used in antitank grenades. HEAT rounds are less effective against composite armor containing ceramic plates.



ARMOR-PIERCING DISCARDING SABOT

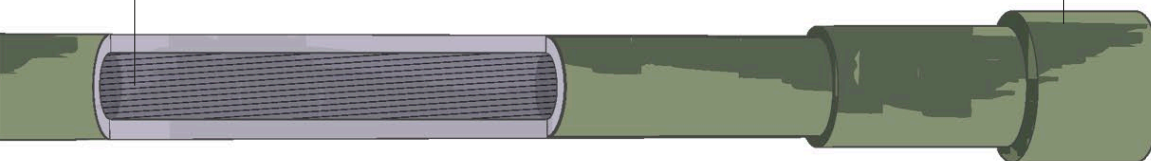
APDS rounds were developed during World War II. Unlike earlier Kinetic Energy rounds, it uses a subcaliber (i.e. smaller than the gun barrel) penetrator encased in a sabot. This design allows for the highest possible velocity, which maximizes armor penetration, combined with the best aerodynamic performance, which ensures high accuracy.



CENTURION MARK 3

Rifling imparts spin to projectiles to increase accuracy

Muzzle counterweight



Ordnance QF 20-pounder

The 20-pounder armed the FV4007 Centurion Mk 3 tank, in service with the British Army (and many others) from 1948. This was a much more powerful weapon than its predecessor, the wartime 17-pounder. It had a caliber of 3,28 in (83.4 mm), and could fire APCBC and APDS antitank rounds, as well as HE, canister, and smoke shells.

SHELL SIZES

The effort to produce increasingly powerful main gun ammunition to counter ever-thicker armor had an entirely predictable effect: the projectiles got bigger, the charge needed to launch them increased proportionately, and so did the length of the cartridge case containing it.

